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Spatially Grounded Surgical Scene Understanding in Minimally Invasive Surgery

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I, Oluwatosin Alabi, confirm that the work presented in this thesis is my own. Where information has been derived from other sources, I confirm that this has been indicated in the thesis.

Abstract

Minimally invasive surgery presents a challenging environment for surgical scene understanding because operative video is visually complex, spatially constrained, and clinically contextual. Recent advances in surgical artificial intelligence have enabled progress in tasks such as instrument segmentation, workflow recognition, and surgical action analysis. However, modelling surgical interactions in a clinically meaningful and interpretable way remains difficult. Surgical action triplets, represented as instrument–verb–target relationships, provide a compact description of surgical activity by specifying which instrument performs which action on which anatomical structure. Existing triplet understanding methods largely operate at the frame level, identifying interactions without explicitly grounding them to the responsible instrument instance.

This thesis advances fine-grained surgical scene understanding by developing datasets, benchmarks, and models for grounded instrument–tissue interaction modelling in laparoscopic surgery. First, it reviews the broader landscape of multitask learning and surgical scene understanding in minimally invasive surgical vision, identifying the progression from perceptual and workflow tasks towards fine-grained action understanding, as well as the lack of explicit spatial grounding for surgical interactions.

Second, this thesis introduces CholecInstanceSeg, a large-scale tool instance segmentation dataset for laparoscopic cholecystectomy. CholecInstanceSeg provides instance-level annotations for surgical instruments across routine clinical procedures, enabling individual tools to be localised and distinguished in complex multi-instrument scenes. This dataset establishes the spatial foundation required for grounded surgical action understanding.

Third, this thesis introduces triplet segmentation, a task that extends surgical action triplet recognition from frame-level prediction to instance-level spatial grounding. To support this task, CholecTriplet-Seg is introduced as a benchmark linking instrument instance masks with verb and anatomical target labels. TargetFusionNet is then proposed as a target-aware architecture that integrates weak anatomical priors into an instance segmentation framework for improved grounded triplet prediction. Experiments show that strong instance supervision substantially improves triplet grounding and that target-aware fusion improves full triplet prediction over existing baselines.

Finally, this thesis investigates whether anatomical target prediction should be approached not only as a representation-learning problem, but also as a reasoning problem. Using multimodal vision-language models and structured target-centric reasoning, it explores how visual evidence, anatomical context, ambiguity analysis, and weak priors can support fine-grained surgical interaction understanding.

Collectively, this work moves surgical action understanding beyond frame-level recognition towards grounded, interpretable, and reasoning-aware modelling of instrument–tissue interactions. These contributions provide foundations for future context-aware surgical assistance, workflow analysis, and intelligent computer-assisted intervention systems.

Impact statement

An impact statement is often required.

Acknowledgments

I'd like to thank my invisible cat.

Publications arising from this thesis

The following publications form the main body of work presented in this thesis.

Journal publications

1. Oluwatosin Alabi, Tom Vercauteren, and Miaoqing Shi. *Multitask Learning in Minimally Invasive Surgical Vision: A Review*. Medical Image Analysis, 101:103480, 2025.
2. Oluwatosin Alabi, Ko Ko Zayar Toe, Zijian Zhou, Charlie Budd, Nicholas Raison, Miaoqing Shi, and Tom Vercauteren. *CholecInstanceSeg: A Tool Instance Segmentation Dataset for Laparoscopic Surgery*. Scientific Data, 12:825, 2025.
3. Oluwatosin Alabi, Meng Wei, Charlie Budd, Tom Vercauteren, and Miaoqing Shi. *Grounding Surgical Action Triplets with Instrument Instance Segmentation: A Dataset and Target-Aware Fusion Approach*. International Journal of Computer Assisted Radiology and Surgery, 2026.

Manuscripts in preparation

1. Oluwatosin Alabi. *From Target Fusion to Target Reasoning: Leveraging Weak Anatomical Priors through Vision-Language Models for Surgical Action Triplet Recognition*. Manuscript in preparation, planned for submission to AAAI.

Relationship to thesis chapters

The review publication forms the basis of Chapter 2. The CholecInstanceSeg publication forms the basis of Chapter 3. The TargetFusionNet and CholecTriplet-Seg publication forms the basis of Chapter 4. The manuscript in preparation motivates the planned work in Chapter 5.

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Introduction

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1.1 Motivation

Minimally invasive surgery (MIS) has transformed surgical care by reducing tissue trauma, blood loss, post-operative pain, recovery time, and complication rates when compared with open surgery [Jaffray 2005]. These benefits arise partly because the operative field is accessed through small incisions and observed through an endoscope or laparoscope. However, the same visual mediation that makes MIS less invasive also makes interpretation more difficult. Surgeons must understand the operative scene from a camera view that is narrow, indirect, dynamic, and frequently degraded by smoke, blood, specular reflections, motion blur, occlusion, tissue deformation, and changing illumination.

Computer-assisted intervention aims to support surgical teams by extracting useful information from these complex visual streams. In the long term, this includes context-aware decision support, surgical workflow monitoring, skill assessment, robotic assistance, automatic camera control, and safer forms of surgical

autonomy. In the short term, it requires a more basic capability: computational systems must be able to understand what is happening in the surgical scene.

Much progress has been made on individual surgical vision tasks. Models can detect tools, segment instruments, recognise surgical phases, estimate depth, and classify activities. Yet surgical understanding is not only a collection of isolated predictions. A clinically meaningful description of a scene often depends on relationships: which instrument is present, where it is, what action it is performing, and which anatomical structure it is interacting with. This thesis is concerned with that relational level of surgical scene understanding.

1.2 From Recognition to Grounded Interaction Understanding

Surgical action triplets provide a compact representation of instrument-tissue interactions. In this formulation, an interaction is described as an `<instrument, verb, target>` tuple, such as `<Grasper, Retract, Gallbladder>` or `<Hook, Dissect, Gallbladder>`. This representation captures more surgical meaning than tool presence or workflow phase alone because it describes an action and its anatomical context.

The CholecT series of datasets, especially CholecT50, has been central to progress in surgical action triplet recognition [Nwoye 2022a]. These datasets and their associated benchmarks have made it possible to compare models on fine-grained action understanding in laparoscopic cholecystectomy. Methods such as TripNet and Rendezvous helped establish the task and demonstrated the value of modelling relationships between instruments, verbs, and targets [Nwoye 2020a; Nwoye 2022a].

However, most surgical action triplet work has historically treated the problem as frame-level recognition. A model predicts that a triplet is present somewhere in the image, but does not necessarily identify which visible instrument instance is responsible for the interaction. This distinction matters. In many laparoscopic frames, multiple instruments are visible, and sometimes multiple instruments of the same class appear simultaneously. If two graspers are present, a frame-level label may say that one grasper retracts the liver and another grasps the gallbladder, but the annotation does not directly specify which grasper performs which action.

This creates a grounding problem. For surgical AI systems to be interpretable and useful, predictions should be linked to the visual evidence that supports them. It is not enough to know that an action occurred somewhere in the frame; the system should indicate the instrument instance involved and the spatial context

of the action. This thesis therefore moves from frame-level recognition towards spatially grounded surgical interaction understanding.

1.3 Research Gap

The work in this thesis was shaped by a review of multitask learning and surgical scene understanding in MIS. That review showed that the field was progressing from coarse workflow analysis towards fine-grained activity understanding, including surgical action triplets. It also showed that many surgical scene understanding tasks are naturally related: instrument recognition, segmentation, phase recognition, action recognition, and target prediction all describe different aspects of the same visual scene.

At the same time, the review exposed two important gaps. First, although triplet recognition had become a meaningful benchmark for surgical activity understanding, the dominant formulation remained weakly grounded or frame-level. Existing methods could often recognise that an interaction was present, but did not strongly link that interaction to a precise instrument instance. Second, public datasets with large-scale instrument instance segmentation annotations in routine human laparoscopic surgery were limited. Without instance-level masks, it is difficult to develop or evaluate strongly supervised models for grounded surgical interaction understanding.

These gaps are connected. Grounded triplet understanding requires both semantic interaction labels and spatial instance annotations. Triplet labels without instance masks leave the actor ambiguous. Instance masks without action and target labels localise instruments but do not describe their surgical role. A central motivation of this thesis is therefore to bring these two forms of supervision together.

A further challenge is anatomical target prediction. Instruments are often visually distinctive, and their masks provide direct spatial support. Anatomical targets are less straightforward. They may have ambiguous boundaries, similar appearance, partial visibility, and definitions that depend on surgical context. A target such as the cystic duct, cystic artery, cystic pedicle, gallbladder, or liver may not be identifiable from local appearance alone. This observation motivates the final part of the thesis, which investigates whether target prediction should be treated not only as a representation-learning problem, but also as a reasoning problem.

1.4 Thesis Aim and Research Questions

The aim of this thesis is to advance surgical scene understanding in minimally invasive surgery by developing datasets, benchmarks, and models for spatially grounded interpretation of instrument-tissue interactions in laparoscopic video.

This aim is addressed through the following research questions:

1. How has multitask learning and related work in surgical scene understanding shaped the progression from perceptual tasks to fine-grained surgical activity understanding?
2. Can a large-scale instrument instance segmentation dataset be constructed for laparoscopic cholecystectomy by extending existing public surgical datasets?
3. Can surgical action triplets be grounded to instrument instance masks, enabling spatially interpretable `<instrument, verb, target>` predictions?
4. Can weak anatomical priors improve target prediction and full triplet grounding when integrated into an instance segmentation architecture?
5. Does the persistent difficulty of target prediction suggest a need for explicit reasoning over anatomy, context, and surgical intent?

1.5 Contributions

The main contributions of this thesis are as follows.

1.5.1 A Review-Based Framing of Surgical Scene Understanding

This thesis first synthesises the literature on multitask learning and surgical scene understanding in MIS. Rather than treating multitask learning as the sole focus, the review is used to position surgical scene understanding as a set of related perceptual, temporal, and semantic tasks. This framing identifies the progression from instrument detection and phase recognition towards fine-grained action triplets, and highlights the lack of strong spatial grounding as a key limitation in existing surgical action understanding.

1.5.2 CholecInstanceSeg

The thesis introduces CholecInstanceSeg, a large-scale open-access dataset for surgical tool instance segmentation in laparoscopic cholecystectomy. The dataset contains 41,933 annotated frames from 85 clinical procedures and 64,483 tool instances.

It provides semantic labels and instance identifiers for laparoscopic instruments, enabling individual tools to be distinguished in multi-instrument scenes.

CholecInstanceSeg addresses the need for instance-level tool annotations in routine human laparoscopic surgery. It also forms the spatial substrate for later work in the thesis, where individual instrument masks are linked to surgical actions and anatomical targets.

1.5.3 CholecTriplet-Seg

Building on CholecInstanceSeg and CholecT50, this thesis introduces CholecTriplet-Seg, a dataset for spatially grounded surgical action triplet understanding. CholecTriplet-Seg links instrument instance masks with verb and anatomical target labels, creating a benchmark for instance-level triplet grounding. The dataset contains 30,955 frames and 49,866 grounded triplets from 50 laparoscopic cholecystectomy videos.

This contribution changes the triplet task from asking which interactions are present in a frame to asking which instrument instance performs each interaction. It therefore enables stronger evaluation of interpretable surgical scene understanding.

1.5.4 TargetFusionNet

The thesis proposes TargetFusionNet, a target-aware architecture for grounded surgical action triplet prediction. TargetFusionNet extends an instance segmentation framework with a gated fusion mechanism that incorporates weak anatomical priors. These priors are not treated as ground truth, but as coarse spatial cues that can help disambiguate instrument-target relationships.

Experiments on CholecTriplet-Seg show that strong instance supervision substantially improves triplet grounding, and that target-aware fusion improves full triplet prediction over weakly grounded and strongly supervised baselines. The results also show that target prediction remains the main bottleneck.

1.5.5 Target Prediction as Reasoning

Finally, the thesis investigates whether anatomical target prediction may require explicit reasoning. This work uses multimodal vision-language models and structured target-centric reasoning to explore whether models can use visual evidence, anatomical context, ambiguity analysis, and weak anatomical priors to improve grounded surgical action triplet recognition.

This contribution reframes target prediction as more than a feature-discrimination problem. It asks whether progress in surgical interaction understanding may de-

pend on models that can reason over surgical context, rather than only learn visual representations.

1.6 Thesis Scope

The thesis focuses on laparoscopic cholecystectomy, a common minimally invasive procedure with established public datasets and benchmarks. This focus allows the work to build on Cholec80, CholecT50, and CholecSeg8k, while maintaining a coherent procedural setting across the dataset and modelling contributions.

The thesis does not attempt to solve all of surgical scene understanding. It does not provide full anatomical segmentation of the surgical scene, nor does it validate a clinical decision-support system prospectively in the operating room. Instead, it addresses a more specific but necessary step: creating and using instance-level supervision to ground surgical interactions in laparoscopic video. The work is intended as a foundation for future systems that combine perception, temporal reasoning, anatomical understanding, and clinical validation.

1.7 Thesis Structure

The remainder of this thesis is organised as follows.

Chapter 2 reviews the literature on surgical scene understanding, surgical action triplets, spatial grounding, multitask learning, datasets, and large models. It positions this thesis within the progression from frame-level recognition towards grounded and reasoning-aware surgical interaction understanding.

Chapter 3 presents CholecInstanceSeg, a large-scale dataset for tool instance segmentation in laparoscopic cholecystectomy. It describes the data sources, annotation protocol, dataset construction, technical validation, and limitations.

Chapter 4 introduces CholecTriplet-Seg and TargetFusionNet. It defines triplet segmentation as a spatially grounded extension of surgical action triplet recognition, presents the aligned dataset, and evaluates target-aware fusion for grounded triplet prediction.

Chapter 5 explores whether anatomical target prediction can be approached as a reasoning problem. It investigates multimodal vision-language models and target-centric reasoning for grounded surgical action triplet recognition.

Chapter 6 synthesises the thesis contributions, discusses limitations, considers ethical and clinical implications, and outlines future work towards richer, more reliable surgical scene understanding.

Background and Literature Review

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Chapter note. This chapter draws on the survey paper *Multitask Learning in Minimally Invasive Surgical Vision: A Review*. The aim here is not to reproduce the survey in full, but to use it as a foundation for the thesis: surgical scene understanding, action triplets, spatial grounding, multitask learning, datasets, and large models.

2.1 Surgical Scene Understanding

Minimally invasive surgery (MIS) has become widely adopted because it can reduce blood loss, post-operative pain, recovery time, and surgical complications when compared with open surgery [Jaffray 2005]. However, the same operating style that makes MIS less invasive also makes visual interpretation more difficult. The surgeon sees the operative field through a laparoscope or endoscope, with a restricted field of view, indirect camera motion, changing illumination, smoke, blood, specular reflection, soft-tissue deformation, and frequent occlusions.

Surgical scene understanding refers to computational methods that infer clinically meaningful information from this visual stream. The field includes low-level

and mid-level perceptual tasks such as instrument detection, instrument segmentation, tissue segmentation, depth estimation, motion estimation, and tracking. It also includes higher-level semantic tasks such as phase recognition, action recognition, action triplet recognition, skill assessment, report generation, and visual question answering. These tasks are not independent: recognising an action often depends on identifying the instrument, localising it, understanding its relation to tissue, and interpreting its temporal context.



Figure 2.1: Application areas in surgical scene understanding covered by the review paper. The categories include perceptual tasks, tracking and control, workflow analysis, anticipation, skill assessment, report generation, and large models.

The literature can be viewed as moving along a hierarchy. At the perceptual level, models extract spatial information from the image. Instrument detection and segmentation locate tools; depth and motion estimation provide geometric cues; tissue segmentation estimates anatomical regions. At the workflow level, models infer progress through the operation, for example by recognising phases, steps, gestures, or activities. At the interaction level, models attempt to describe what

is happening between instruments and tissue. This final level is the focus of this thesis.

2.2 Perceptual Foundations: Detection, Segmentation, and Tracking

Instrument detection and segmentation are foundational because instruments are the most visually salient agents in many laparoscopic scenes. Detection methods localise instruments with bounding boxes, while segmentation methods provide pixel-level support. Semantic segmentation assigns a class to each tool pixel but does not distinguish between multiple tools of the same class. Instance segmentation, by contrast, separates individual instruments and is therefore better suited for interaction understanding.

Early work in surgical instrument segmentation often focused on binary or semantic segmentation, sometimes in porcine or ex-vivo settings. These datasets and methods were valuable for establishing the feasibility of surgical tool localisation, but they were not sufficient for scenes in which multiple instruments of the same type appear simultaneously. This limitation becomes especially important when the goal is to determine which instrument is performing which action.

Tracking and control methods build on detection and segmentation by estimating motion, pose, or saliency over time. Such methods are relevant to automatic camera control and robotic assistance, where the system must decide what part of the scene should remain visible. They also highlight a broader theme: surgical understanding is not only a matter of recognising categories, but of maintaining spatial and temporal consistency.

2.3 Workflow Analysis and Activity Recognition

Surgical workflow analysis studies the temporal structure of operations. Phase recognition has been a central benchmark task, especially in laparoscopic cholecystectomy, where procedures can be decomposed into clinically meaningful stages. Models such as EndoNet helped popularise joint instrument and phase recognition, showing that instrument presence provides useful context for understanding surgical progress [Twinanda 2016].

Phase recognition, however, is coarse. It tells us where we are in the operation, but not what specific interaction is occurring in a frame. More fine-grained activity representations include steps, gestures, actions, and instrument-tissue interactions. These representations are closer to the level at which surgical safety and assistance

become meaningful. For example, a system that recognises only that the surgery is in a dissection phase may be less useful than one that recognises which instrument is dissecting which anatomical structure.

This progression from phase recognition to fine-grained activity recognition was one of the important patterns identified during the review. The field was moving from global descriptors of the procedure towards local descriptions of surgical activity. That observation helped motivate the later focus of this thesis: if surgical activity is represented through instrument-tissue interactions, then those interactions should ideally be spatially grounded in the image.

2.4 Surgical Action Triplets

Surgical action triplets formalise instrument-tissue interaction as a tuple consisting of an instrument, a verb, and an anatomical target. A triplet such as **<Grasper, Retract, Gallbladder>** describes which instrument is acting, what it is doing, and what structure it is acting upon. This representation is more informative than instrument presence or phase recognition because it captures the relational structure of the surgical scene.

Triplet recognition was developed through the CholecT series of datasets. CholecT50 is the most established benchmark, containing 50 laparoscopic cholecystectomy procedures annotated with action triplets [Nwoye 2022a]. It extends earlier CholecT datasets and provides a standard evaluation protocol for component-level and triplet-level average precision. Metrics are typically reported for instruments, verbs, targets, instrument-verb pairs, instrument-target pairs, and full instrument-verb-target triplets.

TripNet introduced one of the early deep learning approaches to surgical action triplet recognition [Nwoye 2020a]. It treated triplet prediction as a structured multilabel problem, with instrument, verb, and target components connected through a valid triplet space. Rendezvous later introduced a more advanced attention-based architecture for triplet recognition, using interacting branches to model instrument, verb, and target relationships [Nwoye 2022a]. Subsequent work explored temporal attention, self-distillation, disentangled classification, and related strategies for improving triplet recognition.

Despite this progress, the dominant formulation remained frame-level recognition. A model could predict that a triplet was present in an image without identifying which visible instrument instance performed that action. This limitation is not just a technical detail. In multi-instrument scenes, especially when multiple instruments share the same class, frame-level labels create a data association problem.

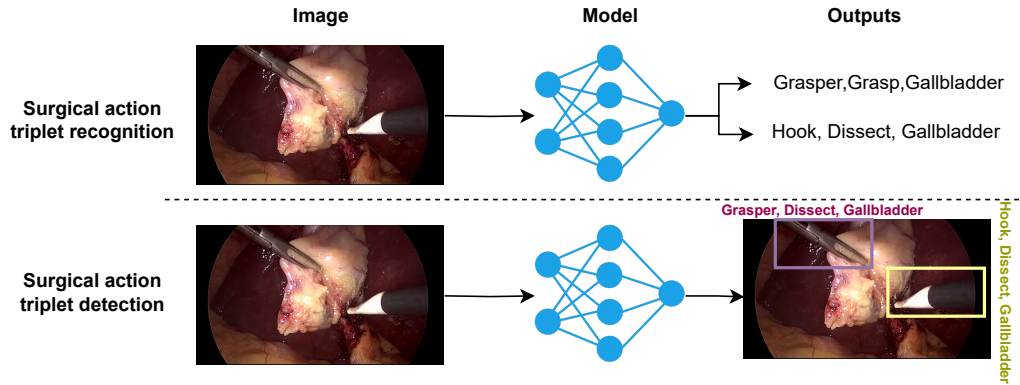


Figure 2.2: Progression of the surgical action triplet task from frame-level triplet recognition towards spatially localised triplet detection. This thesis extends that trajectory towards instance-mask-based triplet grounding.

The interaction is known at the image level but not grounded to a specific actor.

2.5 Surgical Grounding

Grounding refers to linking a semantic prediction to its spatial evidence in the image. In surgical action understanding, grounding asks not only whether a triplet is present, but where it occurs and which instrument instance is responsible for it. This is essential for interpretability: a grounded prediction can be inspected, corrected, and related to surgical anatomy.

Early grounding in triplet recognition relied on weak spatial cues such as class activation maps (CAMs). CAMs can indicate discriminative image regions for a class, but they do not provide precise instance localisation. They are especially limited for laparoscopic instruments, which are long, thin, partially occluded, and often visually entangled with tissue. Bounding-box triplet detection improved the spatial formulation by requiring localisation, but bounding boxes remain coarse for non-axis-aligned instruments.

The review highlighted a gap between the maturity of action triplet recognition and the availability of instance-level tool annotations. At the time the triplet literature was developing, few public surgical datasets provided large-scale instrument instance segmentation in routine human laparoscopic surgery. This made it difficult to study strongly supervised, pixel-level grounding of triplets. The thesis work in Chapter 3 and Chapter 4 responds directly to this gap: first by creating the instance segmentation substrate, then by using it to ground triplet predictions.

2.6 Multitask Learning as a Lens

Multitask learning (MTL) is relevant to this thesis because surgical scene understanding naturally decomposes into related tasks. Instrument recognition, segmentation, phase recognition, action recognition, target prediction, and interaction modelling are not independent problems. They describe different aspects of the same surgical scene. Learning them jointly can improve efficiency, share useful representations, and expose relationships between tasks.

The review found that MTL has been applied across many MIS settings, including perceptual tasks, tracking and control, workflow analysis, anticipation, skill assessment, report generation, and large-model prompting. However, for the purposes of this thesis, the key point is narrower: MTL provides a conceptual language for thinking about how different supervision signals interact. Surgical action triplets are inherently multitask-like because their components are coupled. Instrument, verb, and target prediction can be separated for analysis, but a clinically meaningful interaction requires the three to be predicted coherently.

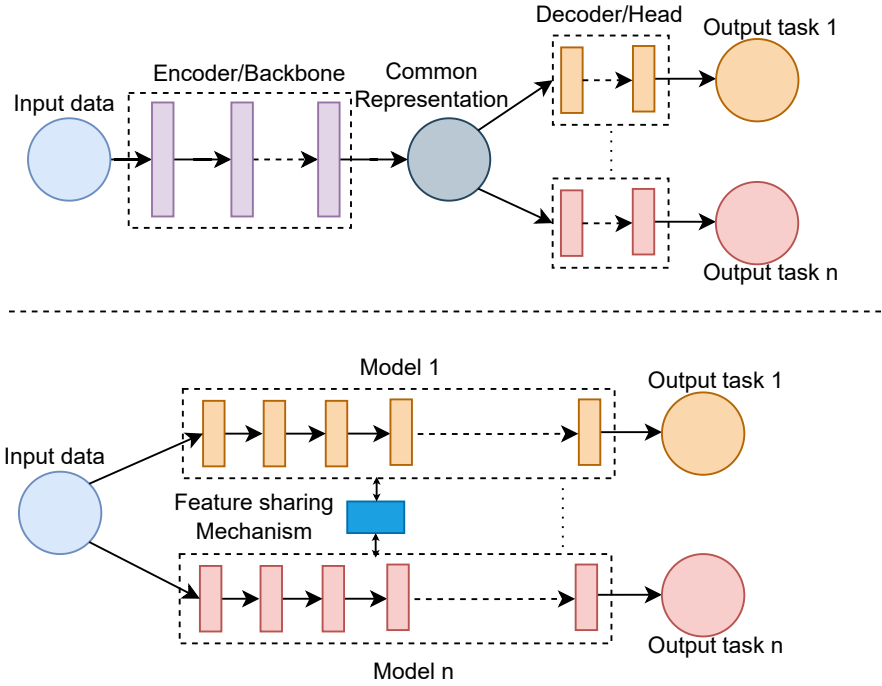


Figure 2.3: Common parameter-sharing strategies in multitask learning. In surgical scene understanding, hard parameter sharing with a shared encoder and task-specific heads has been especially common.

In general computer vision, MTL has explored hard parameter sharing, soft parameter sharing, task balancing, auxiliary objectives, and data-efficient learning.

In MIS, however, the review found that many methods rely on relatively simple hard-parameter-sharing architectures and linear combinations of task losses. More advanced task-balancing or optimisation methods are less common. This matters because surgical tasks can interfere with one another: improving one component may not improve the full interaction, and a model that predicts instruments and targets well separately may still fail to produce coherent triplets.

For this thesis, MTL is therefore not the main subject, but a useful diagnostic lens. It helps explain why interaction understanding is difficult: the model must coordinate multiple related predictions under limited and imbalanced supervision. This perspective reappears in Chapter 4, where a 100-class triplet formulation is compared with decomposed multitask alternatives.

2.7 Datasets for Surgical Scene Understanding

Progress in surgical scene understanding is closely tied to dataset availability. Cholec80 enabled large-scale work on phase recognition and tool presence. CholecT50 enabled action triplet recognition. CholecSeg8k provided semantic segmentation labels for selected cholecystectomy frames. Other datasets support tasks such as workflow analysis, surgical skill assessment, VQA, and multi-granularity activity recognition.

However, the review also made clear that datasets are often fragmented by task. A dataset may provide phase labels without pixel-level tool masks, or segmentation labels without action triplets, or triplets without instance-level grounding. This fragmentation limits the kinds of questions that can be studied. For example, CholecT50 supports triplet recognition, but its triplets are frame-level. CholecSeg8k supports semantic segmentation, but it does not provide instance segmentation for the full range of tools. The absence of aligned annotations across tasks makes it difficult to train models that connect perception, action, and anatomy.

This dataset gap is central to the thesis. Chapter 3 constructs a large-scale instance segmentation dataset by building on Cholec80, CholecT50, and CholecSeg8k. Chapter 4 then aligns instrument instances with CholecT50 triplets to create CholecTriplet-Seg. In this way, the thesis follows a data-first strategy: before a model can reason about grounded interactions, the relevant visual entities must be annotated and linked.

2.8 Vision-Language Models and Large Models in Surgery

The review also identified the growing role of large models in surgical scene understanding. Large pretrained models such as language models, vision-language mod-

els, and promptable segmentation models are attractive because they can support multiple tasks through prompting or adaptation. In surgery, this has led to work on surgical visual question answering, report generation, promptable segmentation, and general-purpose surgical assistants.

Surgical VQA systems attempt to answer natural-language questions about images or videos, including questions about instruments, tissues, actions, and scene context. Promptable segmentation methods attempt to adapt large segmentation or vision-language models to surgical instruments and tissues. These approaches are still limited by domain shift, data scarcity, real-time deployment constraints, and the need for clinically reliable outputs. Nevertheless, they suggest an important direction: some surgical understanding tasks may require models that can combine perception with explicit reasoning.

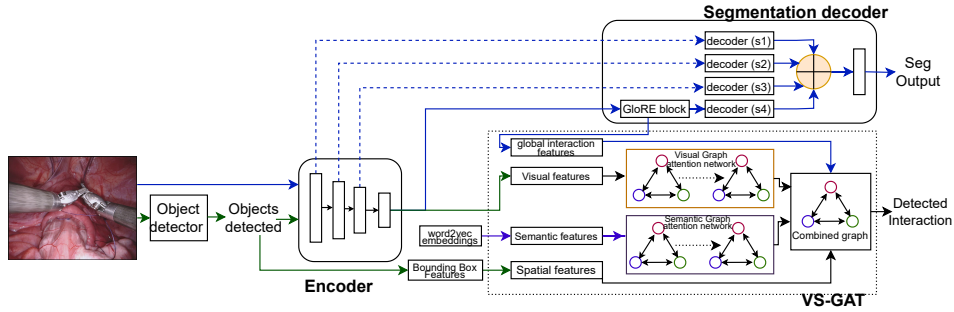


Figure 2.4: Example of a large-model-style formulation for surgical scene understanding, where a model is prompted to solve multiple related tasks. Such models motivate the later question of whether target prediction can benefit from explicit reasoning.

This observation becomes important in Chapter 5. The target component of surgical action triplets is often the weakest part of triplet prediction. Unlike instruments, anatomical targets may not have clear boundaries. Unlike verbs, they may not be inferable from tool pose alone. Human observers often rely on anatomical context, procedural intent, and ambiguity resolution. Vision-language models offer a way to test whether explicit reasoning over such context can improve target prediction.

2.9 Thesis Positioning

The literature reviewed in this chapter suggests a clear trajectory. Surgical scene understanding has progressed from tool presence and phase recognition towards finer-grained representations of activity. Surgical action triplets provide a compact

and clinically meaningful representation of instrument-tissue interactions. However, most triplet methods have historically stopped at frame-level recognition, leaving the interaction weakly grounded or ungrounded in the image.

The review process also exposed a practical bottleneck: grounding triplets requires reliable instrument instance annotations, but large-scale public instance segmentation datasets for routine laparoscopic surgery were limited. This gap motivated the dataset contribution in Chapter 3. Once instrument instances were available, it became possible to ask a stronger question: can each surgical action triplet be linked to the exact instrument instance that performs it? This question motivates CholecTriplet-Seg and TargetFusionNet in Chapter 4.

Finally, the literature on large models and surgical VQA points towards a further step. Even with spatial grounding, target prediction remains difficult because anatomical interpretation is often contextual and ambiguous. This thesis therefore ends by investigating whether target prediction can be treated not only as a representation-learning problem, but also as a reasoning problem.

In summary, this thesis is positioned at the intersection of surgical scene understanding, action triplet recognition, instance segmentation, and emerging reasoning-based models. Its central argument is that useful surgical interaction understanding requires moving beyond frame-level recognition towards spatially grounded and context-aware interpretation of instrument-tissue interactions.

CholecInstanceSeg: Tool Instance Segmentation for Laparoscopic Surgery

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Chapter note. This chapter adapts material from the Scientific Data paper *CholecInstanceSeg: A Tool Instance Segmentation Dataset for Laparoscopic Surgery*. The chapter is written as the dataset foundation for the subsequent work on grounded surgical action triplets.

3.1 Overview

The previous chapters motivate surgical scene understanding as a central problem in computer-assisted intervention. A recurring limitation in this area is that many datasets describe the presence of tools or provide semantic tool regions, but do not distinguish between individual tool instances. This distinction matters in laparoscopic surgery because multiple instruments of the same class may appear in the same frame and may participate in different actions. A model that only knows that a grasper is present cannot reliably determine which grasper is retracting tissue, which one is idle, or which one is interacting with a particular anatomical target.

This chapter presents CholecInstanceSeg, a large-scale dataset for surgical tool instance segmentation in laparoscopic cholecystectomy. The dataset contains 41,933 annotated frames from 85 clinical procedures and 64,483 tool instances. Each visible surgical instrument is annotated with a semantic class and an instance identifier, enabling pixel-level localisation and separation of individual tools. The annotated classes are grasper, bipolar, hook, clipper, scissors, irrigator, and snare.

The role of this chapter in the thesis is twofold. First, it establishes a high-quality data resource for instrument instance segmentation in routine human laparoscopic surgery. Second, it provides the spatial representation needed by Chapter 4, where instrument instances are linked to surgical action triplets.

3.2 Motivation

Minimally invasive surgery relies on indirect vision through a laparoscope or endoscope. This creates a constrained and visually challenging operating environment, with a limited field of view, frequent occlusion, smoke, specular reflection, motion blur, dirty lenses, and visually similar tool shafts. Computer-assisted intervention systems must therefore interpret scenes from image evidence that is often ambiguous and incomplete.

Tool localisation is one of the most basic requirements for such systems. It supports downstream tasks including workflow analysis, tool-tissue interaction recognition, surgical skill assessment, camera control, and context-aware assistance. However, many public surgical tool datasets are limited in one or more ways: they focus on porcine or ex-vivo settings, contain a small number of annotations, provide only semantic segmentation, or offer instance information for only a restricted set of tool classes. These limitations restrict the development of models that need to reason about individual instrument instances.

Dataset	Patient	Environment	Procedures	Annotations	Semantic	Instance
Endovis2015 [Bodenstedt 2018]	Human	Ex-vivo/In-vivo	8	10k	3 classes	No
Endovis2017 [Allan 2019]	Porcine	In-vivo	10	2.4k	10 classes	No
Endovis2018 [Allan 2020]	Porcine	In-vivo	16	2.4k	10 classes	No
CholecSeg8k [Hong 2020b]	Human	In-vivo	17	8k	12 classes	No
AutoLaparo [Wang 2022]	Human	In-vivo	21	1.8k	7 classes	No
ROBUSTMIS2019 [Ross 2020]	Human	In-vivo	30	10k	2 classes	Yes
SAR-RARP50 [Psychogyios 2023]	Human	In-vivo	50	10k	10 classes	No
CholecInstanceSeg	Human	In-vivo	85	41.9k	8 classes	Yes

Table 3.1: Comparison of representative datasets used in surgical tool segmentation research. CholecInstanceSeg provides instance-level annotations at a larger scale than the listed alternatives while focusing on routine human laparoscopic cholecystectomy.

CholecInstanceSeg was designed to address this gap by expanding existing cholecystectomy datasets with instance segmentation labels. The resulting resource is not only useful for training instance segmentation models, but also for integrating tool localisation with other annotations derived from the same parent videos, such as phase labels, action triplets, and critical view of safety assessment.

3.3 Data Sources

CholecInstanceSeg was created from frames extracted from three public laparoscopic cholecystectomy datasets: Cholec80 [Twinanda 2016], CholecT50 [Nwoye 2022a], and CholecSeg8k [Hong 2020b]. These datasets are public subsets of the CAMMA Cholec120 collection [Twinanda 2018]. Their relationship is shown in Fig. 3.1.

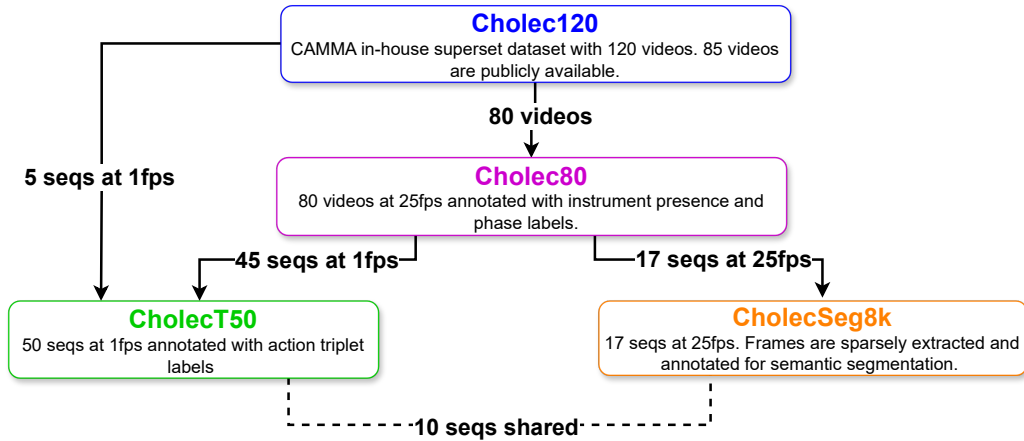


Figure 3.1: Relationship between the source datasets used to construct CholecInstanceSeg. Cholec80, CholecT50, and CholecSeg8k are public subsets of the larger Cholec120 collection. Some image sequences are shared between CholecT50 and CholecSeg8k, although their frame extraction protocols differ.

Cholec80 contains 80 cholecystectomy videos with phase and tool-presence labels. CholecT50 contains 50 image sequences annotated with instrument-action-target triplets. CholecSeg8k contains 8,080 frames with semantic segmentation labels. The overlap between these resources makes them valuable for building multi-annotation surgical datasets, but it also requires careful accounting of shared sequences, frame extraction rates, and potential redundancy across partitions.

The frames from Cholec80 and CholecT50 are primarily at 854 by 480 pixels, with three CholecT50 sequences available at 1920 by 1080 pixels. CholecT50 provides pre-extracted image sequences at 1 frame per second. For Cholec80 videos

not covered by CholecT50 or CholecSeg8k, frames were extracted at a sparse rate of one frame every 30 seconds of the original 25 fps video stream, with additional frames included when a sequence would otherwise contain fewer than 50 sampled frames.

3.4 Dataset Construction

The final dataset is organised into four partitions, shown in Fig. 3.2. These partitions balance annotation density and procedure diversity.

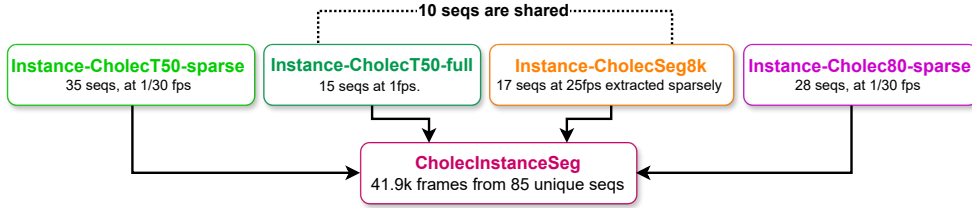


Figure 3.2: Composition of CholecInstanceSeg. The dataset combines dense and sparse partitions from CholecT50, CholecSeg8k, and Cholec80. The mixed strategy provides dense supervision for selected sequences while preserving broad coverage across procedures.

	Inst-CholecSeg8k	Inst-CholecT50-sparse	Inst-CholecT50-full	Inst-Cholec80-sparse	Total
Sequences	17 (10 shared)	35	15 (10 shared)	28	85
Frames	8,080	2,681	28,317	2,855	41,933
Tool instances	10,523	4,098	45,221	4,641	64,483

Table 3.2: Summary of the CholecInstanceSeg partitions.

The Instance-CholecSeg8k partition contains all 8,080 frames from CholecSeg8k. Since CholecSeg8k already provided semantic segmentation, this partition served as a useful starting point, but it required conversion to instance segmentation and further correction. The Instance-CholecT50-full partition contains all frames from 15 selected CholecT50 sequences, giving dense supervision for a subset of procedures. The remaining 35 CholecT50 sequences were sampled sparsely to form Instance-CholecT50-sparse. Finally, 28 Cholec80 videos not included in CholecT50 or CholecSeg8k were sampled sparsely to increase procedural diversity.

This mixed design reflects a practical trade-off. Dense instance segmentation of every frame in every sequence would be prohibitively expensive. At the same time, relying only on sparse annotation would limit temporal and scene coverage. CholecInstanceSeg therefore combines dense annotation where it is most useful with sparse annotation across a wider set of procedures.

3.5 Annotation Protocol

The annotation protocol was designed to support consistent instance-level labelling of laparoscopic instruments. The seven annotated tool categories were:

- grasper,
- bipolar,
- hook,
- clipper,
- scissors,
- irrigator,
- snare.

The snare category was included in addition to the six standard Cholec80/CholecT50 tool categories because it has distinct visual characteristics and a distinct surgical function. Objects such as detached clips, sutures, needles, specimen bags, gauze, and camera ports were excluded. Instrument ports were not annotated when no instrument extended from them; when an instrument visibly extended from a port, the visible port segment was annotated as part of the instrument, following the convention used in CholecSeg8k.

The most important annotation principle was to capture every visible instrument instance while preserving the functional shape of the tool, especially the end-effector. Pixel-perfect outlines are unrealistic at this scale, particularly in difficult frames, but the annotations were required to preserve the identity and spatial extent of each instrument. Holes within tools were treated as part of the tool mask, following common conventions in surgical instrument segmentation datasets.

Routine surgical video contains many hard cases. CholecInstanceSeg explicitly accounts for motion blur, smoke, occlusion by tissue or blood, tissue attached to instruments, instruments near the image edge, instruments far from the camera, specular reflection, low lighting, saturated lighting, dirty lenses, camera-port views, and instruments submerged in fluid. Examples of these challenges are shown in Fig. 3.3.

3.6 Annotation Tools and Workflow

Annotation was performed using a lightly customised version of the Segment Anything Annotator tool [Wang 2023], built on LabelMe [Wada 2016] and the Segment

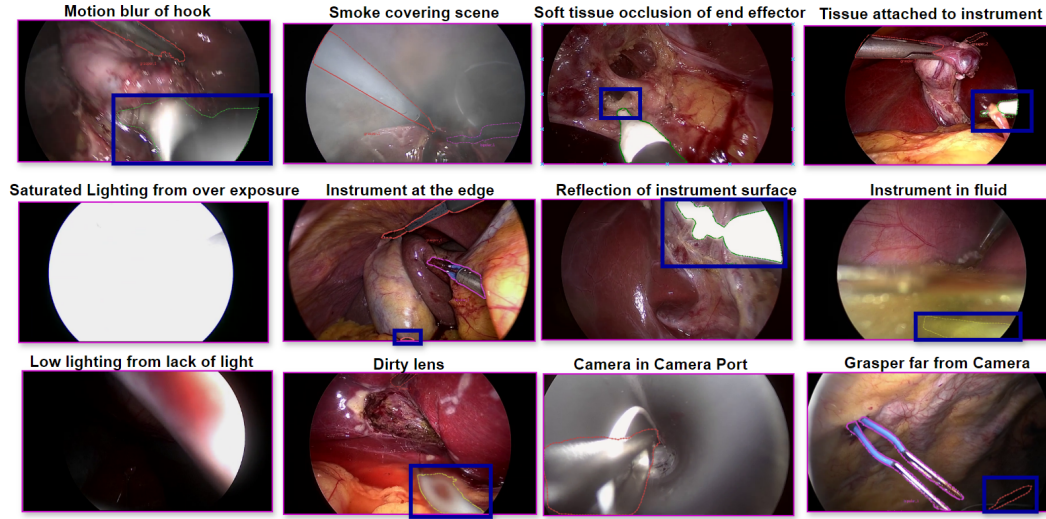


Figure 3.3: Examples of challenging annotation cases in CholecInstanceSeg, including motion blur, smoke, soft-tissue occlusion, tissue attachment, lighting artefacts, reflections, dirty lenses, fluid, and edge-of-frame instruments.

Anything Model (SAM) [Kirillov 2023]. The tool was customised to improve navigation, reduce label-file size, and support quality-control workflows. Point-prompt interaction with SAM was used when possible, while manual polygon correction was used in cases where interactive segmentation failed because of smoke, blur, reflection, low contrast, or visually similar background.

For CholecSeg8k, the existing semantic masks were converted into instance masks using class-aware connected components. This approach separates disconnected instruments of the same class and naturally separates instruments of different classes. However, it can fail when the same instrument is split by occlusion, when two instruments of the same class overlap, or when the source semantic mask contains noise. A total of 892 frames with such issues were identified and corrected.

The dense CholecT50-full partition was created with a human-in-the-loop semi-automatic pipeline, shown in Fig. 3.4. Initial annotations from Instance-CholecSeg8k and Instance-CholecT50-sparse were used to train an RTMDet-Ins-l model [Lyu 2022]. The model generated annotation proposals for unlabelled frames, human annotators corrected the proposals, and the corrected labels were then used to retrain the model. This cycle allowed the dataset to scale while retaining human oversight.

The overall workflow proceeded in stages: conversion of CholecSeg8k to instance labels, development of the labelling protocol, annotation of the sparse CholecT50 partition, semi-automatic annotation of the dense CholecT50 partition, sparse annotation of additional Cholec80 videos, and final quality control. The primary

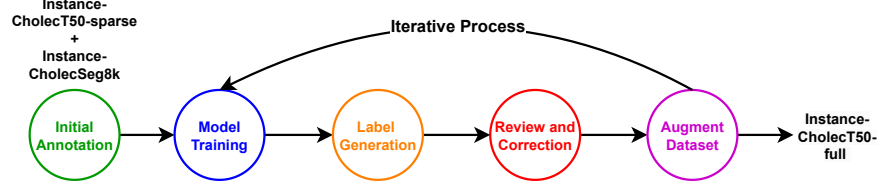


Figure 3.4: Human-in-the-loop semi-automatic annotation pipeline used for CholecInstanceSeg. Model-generated proposals were reviewed and corrected by annotators, then used to expand the training data for subsequent annotation rounds.

annotator performed most annotation and quality-control work; a secondary annotator with medical experience supported review, and an expert team mediated ambiguous cases.

3.7 Data Records and Splits

Annotations follow the LabelMe JSON format. Each annotated shape contains a class label, polygon points, and a group identifier corresponding to the instance ID. When an instrument is represented by multiple polygons, the polygons share a group identifier so that they are treated as a single instance. Figure 3.5 summarises the released folder structure and annotation format.

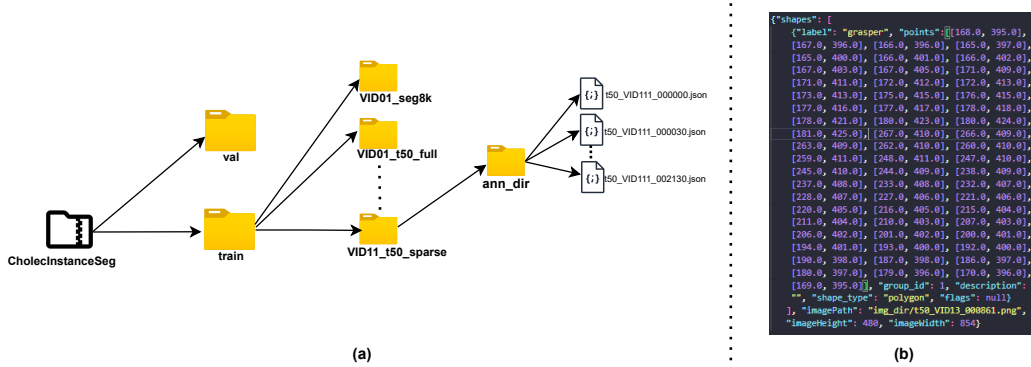


Figure 3.5: Dataset directory structure and JSON annotation format for CholecInstanceSeg. The format follows LabelMe conventions, with class labels, polygon coordinates, and group identifiers used to recover instrument instances.

The recommended split contains 55 training sequences, 17 validation sequences, and 23 test sequences. These splits were designed to account for shared sequences across source datasets, to avoid redundancy between validation/test sets, and to preserve diversity across the evaluation data. The split contains 26,830 training frames, 4,284 validation frames, and 10,819 test frames.

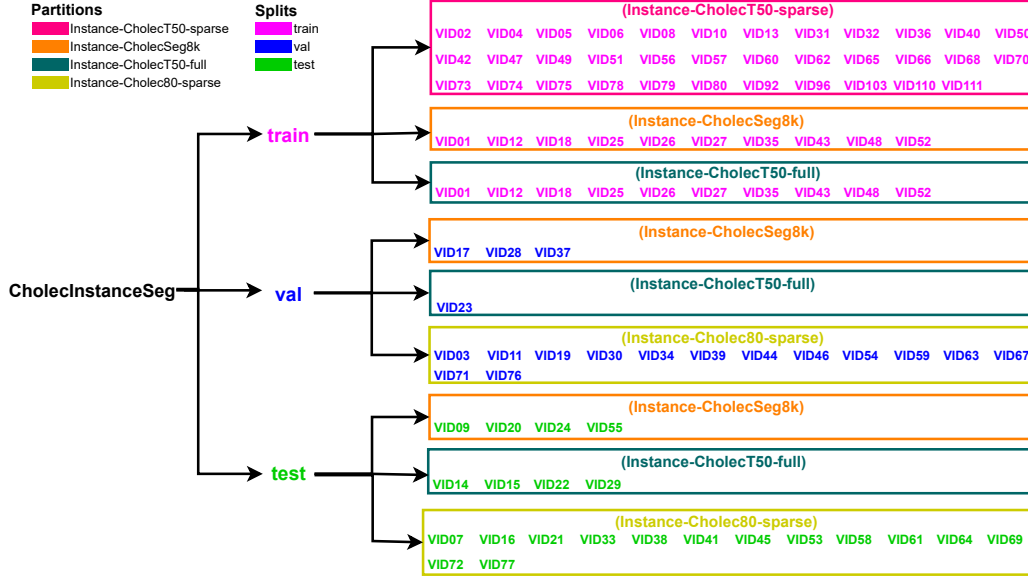


Figure 3.6: Recommended training, validation, and test splits for CholecInstanceSeg, accounting for dataset partitions and shared sequences.

3.8 Dataset Analysis

The class distribution in CholecInstanceSeg is imbalanced, reflecting the realities of laparoscopic cholecystectomy. Graspers are the most frequent instruments, followed by hooks. Irrigators, clippers, scissors, bipolars, and snares occur less often. The number of tools per frame is also uneven: most frames contain one or two visible instruments, with a maximum of four tools in a single frame. Approximately 5,328 frames contain no visible tools.

Figure 3.8 shows how tool instances are distributed across sequences and partitions. The dense CholecT50-full partition contains the largest number of instances, while the sparse partitions provide additional diversity with fewer frames per sequence.

3.9 Technical Validation

CholecInstanceSeg was validated through baseline model training, label-agreement analysis, and quality-control procedures. Two general-purpose instance segmentation models were trained as baselines: Mask R-CNN [He 2017] and Mask2Former [Cheng 2022a]. These were chosen as broadly applicable reference models rather than specialised surgical-tool architectures, so that future work can compare against a clear baseline.

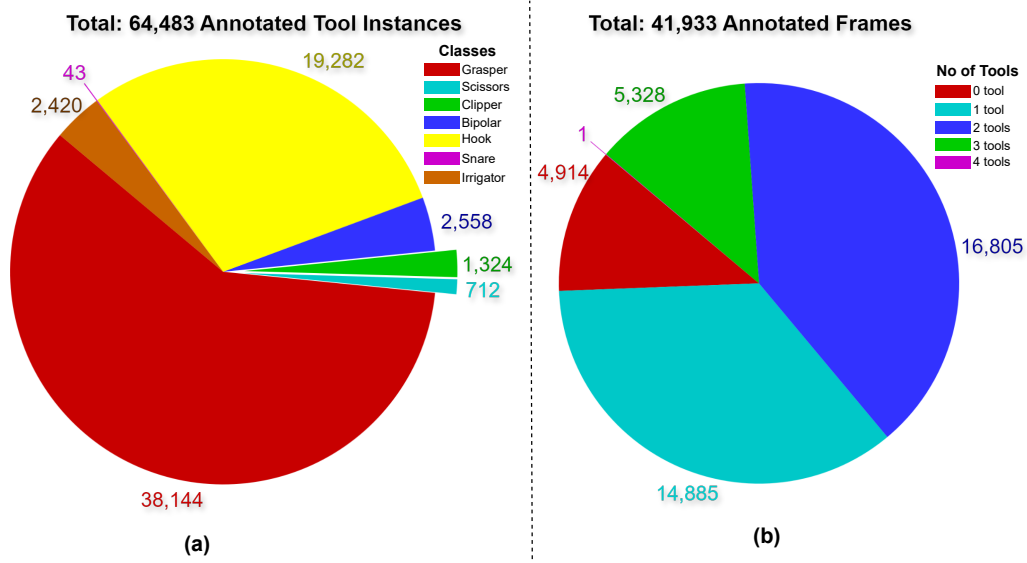


Figure 3.7: Tool class distribution and number of tool instances per frame in CholecInstanceSeg.

Performance was evaluated using COCO mean average precision (mAP) and a sequence-aware variant, sequence mAP (smAP). The standard mAP captures class-level segmentation quality, while smAP averages per-sequence performance to better reflect generalisation across surgical scenes. The baseline results are shown in Table 3.3 and Table 3.4.

Method	GR	HO	IR	CL	BI	SC	SN	mAP
Mask R-CNN [He 2017]	37.25	61.84	43.04	58.73	28.42	20.91	-	41.70
Mask2Former [Cheng 2022a]	61.10	82.16	68.38	81.30	48.02	27.89	-	61.47

Table 3.3: Baseline instance segmentation performance on CholecInstanceSeg using mAP. GR: grasper; HO: hook; IR: irrigator; CL: clipper; BI: bipolar; SC: scissors; SN: snare.

Method	GR	HO	IR	CL	BI	SC	SN	smAP
Mask R-CNN	41.65 $\pm$ 12.74	64.06 $\pm$ 9.20	52.50 $\pm$ 20.09	64.85 $\pm$ 16.20	37.62 $\pm$ 20.74	40.38 $\pm$ 23.54	-	50.18
Mask2Former	66.46 $\pm$ 8.52	88.90 $\pm$ 7.58	74.51 $\pm$ 20.35	84.69 $\pm$ 16.88	65.20 $\pm$ 25.36	69.76 $\pm$ 33.20	-	74.92

Table 3.4: Baseline instance segmentation performance using sequence mAP. The standard deviation indicates variability across sequences.

Mask2Former substantially outperformed Mask R-CNN on both metrics. Hook and clipper achieved high performance, while scissors and bipolar were more challenging, reflecting both lower frequency and visual variability. The smAP results also show that performance can vary considerably across surgical sequences, espe-

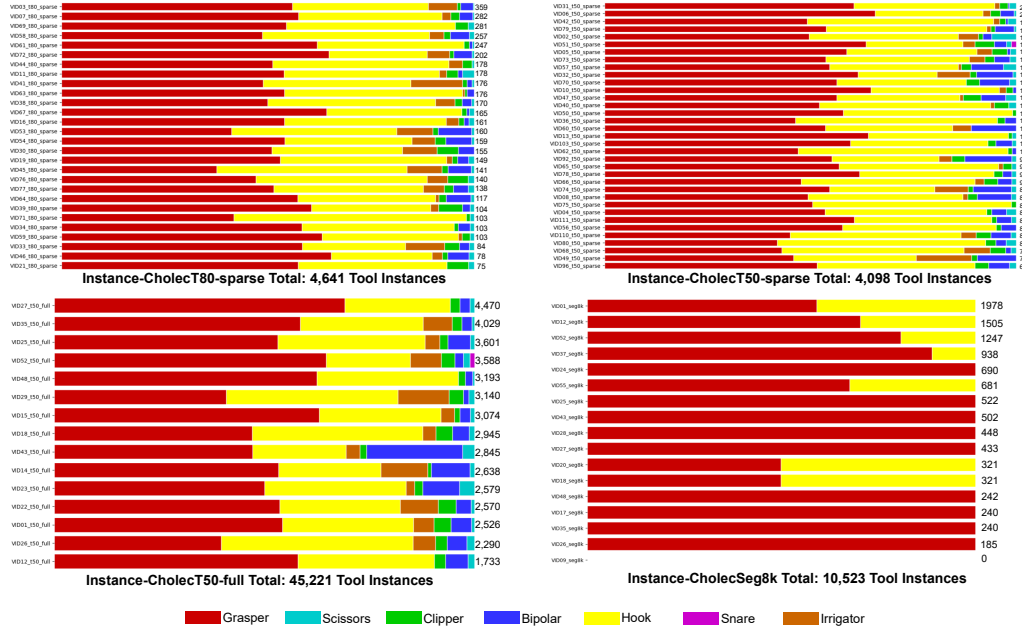


Figure 3.8: Distribution of annotated tool instances across sequences and dataset partitions.

cially for less common tools.

Inter-annotator agreement was measured on a subset of independently annotated data using panoptic quality [Kirillov 2019]. The agreement score was 91.2, indicating strong consistency. Agreement between manual annotations and the semi-automatic annotation pipeline was also measured using panoptic quality, achieving 95.7. These results support the reliability of the annotation process while acknowledging that surgical video annotation remains inherently difficult.

Quality control combined manual review with dataset-specific checks. Frames with more than three tool instances were reviewed, identifying and correcting 26 noisy frames. Cholec80 tool-presence labels were used to cross-check expected tools, identifying 267 frames with errors. Finally, frames with instrument overlaps greater than an IoU threshold of 0.1 were reviewed, leading to corrections in 247 frames.

3.10 Limitations and Role in the Thesis

CholecInstanceSeg has several limitations. It focuses on laparoscopic cholecystectomy, so generalisation to other procedures requires further validation. The annotations are instance-level tool masks rather than full anatomical scene annotations, so they do not directly solve target localisation or tissue understanding. The dataset

is also class-imbalanced, reflecting the natural distribution of instruments in cholecystectomy and making rare tools harder to model.

Despite these limitations, CholecInstanceSeg provides the spatial foundation needed for the remainder of this thesis. By distinguishing individual instrument instances, it enables the next step: linking each instrument to the action it performs and the anatomical target it interacts with. Chapter 4 builds directly on this dataset to move from instrument localisation to grounded surgical interaction understanding.

Grounding Surgical Action Triplets with Instrument Instance Segmentation

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Chapter note. This chapter adapts material from the manuscript *Grounding Surgical Action Triplets with Instrument Instance Segmentation: A Dataset and Target-Aware Fusion Approach*. It builds directly on CholecInstanceSeg by using instrument instance masks as the spatial support for surgical action triplets.

4.1 Overview

The previous chapter established CholecInstanceSeg as a dataset for distinguishing and segmenting individual surgical instruments. This chapter asks the next question: once an instrument instance has been localised, can its action and anatomical target be predicted in a spatially grounded way?

Surgical action triplets represent instrument-tissue interactions as $\langle \text{instrument}, \text{verb}, \text{target} \rangle$ tuples. For example, $\langle \text{Grasper}, \text{Grasp}, \text{Gallbladder} \rangle$ and $\langle \text{Hook}, \text{Dissect}, \text{Gallbladder} \rangle$ describe not only which instrument is present, but also what it is doing and which anatomical structure is involved. This representation is useful because surgical progress and safety are defined by interactions rather than by isolated objects. However, most existing triplet-recognition methods operate at the frame level: they predict which triplets occur somewhere in the image without identifying the instrument instance responsible for each action.

This chapter introduces triplet segmentation, a task that unifies instrument instance segmentation with surgical action triplet prediction. A triplet segmentation model outputs pixel-level instrument masks, and each mask is assigned an instrument, verb, and target label. This makes the prediction spatially grounded and interpretable: each predicted interaction is tied to a visible instrument instance.

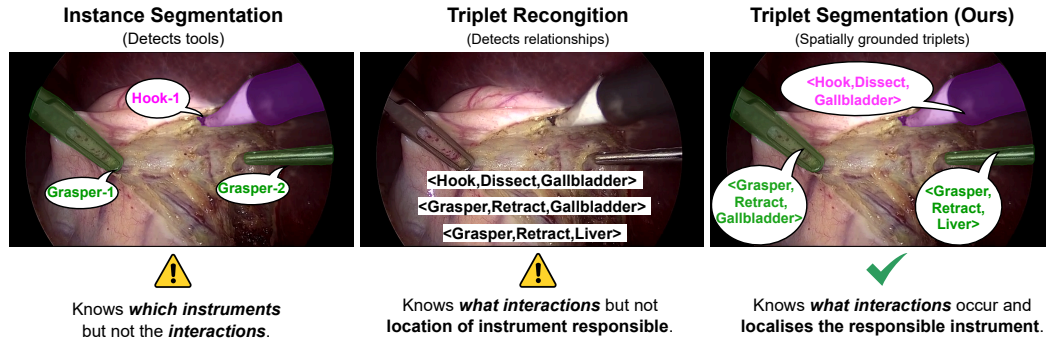


Figure 4.1: Comparison of instance segmentation, frame-level triplet recognition, and triplet segmentation. Triplet segmentation grounds each $\langle \text{instrument}, \text{verb}, \text{target} \rangle$ prediction to an instrument instance mask, making the interaction spatially interpretable.

The chapter makes four contributions. First, it defines triplet segmentation as a spatially grounded surgical action understanding task. Second, it presents CholecTriplet-Seg, a dataset that links CholecInstanceSeg instrument masks with CholecT50 action triplets. Third, it introduces a triplet segmentation evaluation protocol based on recognition, detection, and segmentation mAP. Fourth, it presents TargetFusionNet, a target-aware model that uses weak anatomical priors to improve triplet prediction.

4.2 From Frame-Level Triplets to Instance-Level Grounding

Triplet recognition was introduced to model fine-grained instrument-tissue interactions in laparoscopic video [Nwoye 2020b; Nwoye 2022b]. In this formulation, an image may contain multiple valid triplets, and the model predicts which triplets are present. While this is a powerful representation, frame-level recognition has a structural limitation: it does not indicate which instrument performs which action.

This limitation becomes important in multi-instrument scenes. If two graspers are visible and the frame-level labels include `<Grasper, Retract, Liver>` and `<Grasper, Grasp, Gallbladder>`, the labels alone do not say which grasper corresponds to which interaction. A model can therefore achieve a correct frame-level prediction while still failing to provide a clinically or visually meaningful explanation of the scene.

Earlier spatial grounding approaches relied primarily on class activation maps or bounding boxes, which provide coarse localisation but do not explicitly link actions to precise instrument instances. This is particularly limiting for laparoscopic tools, which are elongated, thin, often non-axis-aligned, and frequently overlap tissue or other instruments. Pixel-level instance masks are a more suitable spatial representation.

Triplet segmentation addresses this by requiring each predicted triplet to be associated with an instrument instance mask. Given an input image $x \in \mathbb{R}^{H \times W \times 3}$, the model predicts a set of masks $M = \{M_k\}$, where each mask is associated with a triplet:

$$\mathcal{T} = \{\langle I_k, v_k, t_k \rangle \mid M_k \in M\}.$$

Here I_k , v_k , and t_k denote the instrument, verb, and target for instance k . The task therefore evaluates both whether the interaction is classified correctly and whether it is localised to the correct instrument instance.

4.3 CholecTriplet-Seg

CholecTriplet-Seg was constructed by aligning frame-level triplet annotations from CholecT50 [Nwoye 2022b] with instrument instance masks from CholecInstanceSeg [Alabi 2025]. The resulting dataset contains 30,955 frames from 50 laparoscopic cholecystectomy videos, with 49,866 spatially grounded triplets.

The triplet vocabulary follows CholecT50: six instruments, ten verbs, and fifteen

Dataset	Videos	Annotation type	Frames
CholecT50 [Nwoye 2022b]	50	Frame-level triplet labels	100,861
CholecInstanceSeg [Alabi 2025]	85	Instrument instance masks	41,933
CholecTriplet-Seg	50	Triplet + instance masks	30,955

Table 4.1: Datasets involved in constructing CholecTriplet-Seg. CholecTriplet-Seg combines the semantic interaction labels of CholecT50 with the spatial instance masks of CholecInstanceSeg.

anatomical targets. Although these categories produce 900 possible instrument-verb-target combinations, CholecT50 restricts the vocabulary to 100 clinically meaningful triplets. CholecTriplet-Seg adopts this clinically informed vocabulary for comparability with previous work; 93 of the 100 valid triplets appear in the final aligned dataset.

4.3.1 Alignment and Annotation Format

The annotation schema extends the LabelMe-style structure used by CholecInstanceSeg. Each instrument shape already contains a class label, polygon points, and an instance group identifier. CholecTriplet-Seg adds two fields: the verb and the anatomical target. If an instrument instance is represented by multiple polygons, all polygons share the same group identifier, label, verb, and target.

The initial alignment matched CholecT50 triplets to CholecInstanceSeg masks using frame correspondence and instrument class. This automatic process was insufficient in several common cases because CholecT50 labels are frame-level rather than instance-level. Manual verification and correction were therefore required to ensure that each triplet was assigned to the correct instrument mask.

Two types of inconsistencies were especially important:

- **Matching errors:** frames where multiple instruments of the same class were associated with different actions, making automatic assignment ambiguous. These affected 2,361 frames.
- **Conflict errors:** frames where the segmentation and triplet annotations disagreed, such as missing masks, redundant triplets, or mismatched class labels. These affected 7,194 frames.

After alignment, a full dataset pass was performed to enforce schema consistency and verify that each instrument instance had a verb and target assignment. This process produced a strongly supervised benchmark for instance-level triplet grounding.

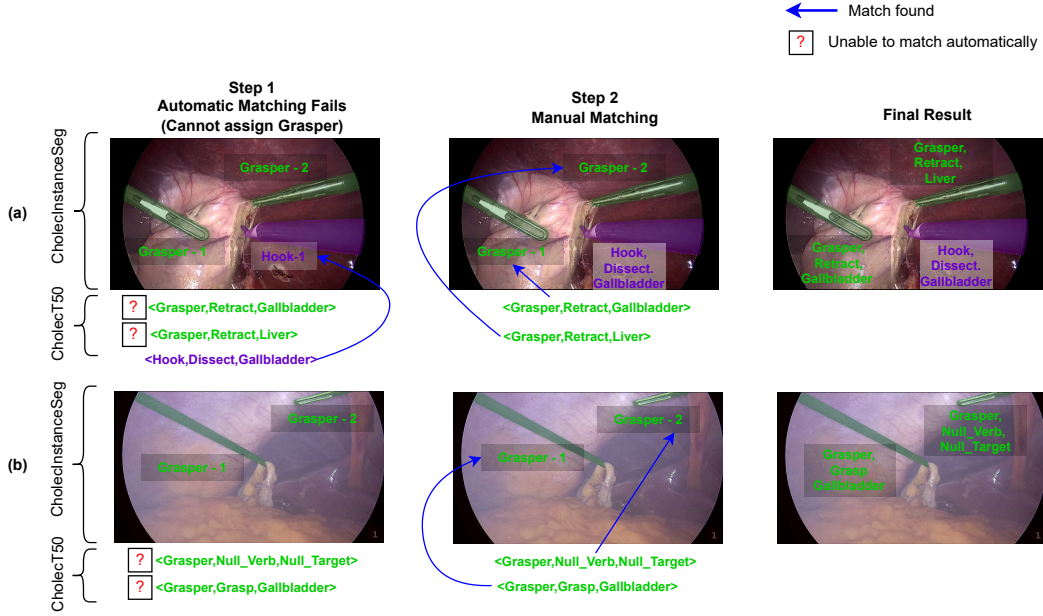


Figure 4.2: Examples of matching errors. Multiple instruments of the same class can perform different actions, so frame-level triplets must be manually linked to the correct instrument instance.

4.3.2 Splits and Class Imbalance

CholecTriplet-Seg follows the CholecInstanceSeg split protocol to avoid leakage between training and evaluation data. The dataset contains 45 training videos, one validation video, and four test videos. The split statistics are shown in Table 4.2.

Split	Videos	Images	Triplets	Avg. triplets/frame
Train	45	22,818	37,591	1.65
Validation	1	2,496	3,341	1.34
Test	4	5,641	8,934	1.58
Total	50	30,955	49,866	1.61

Table 4.2: CholecTriplet-Seg split statistics.

Like many surgical datasets, CholecTriplet-Seg is long-tailed. A small number of common triplets dominate the dataset, while many clinically valid triplets occur rarely. This imbalance is visible both at the full triplet level and within the individual instrument, verb, and target categories.

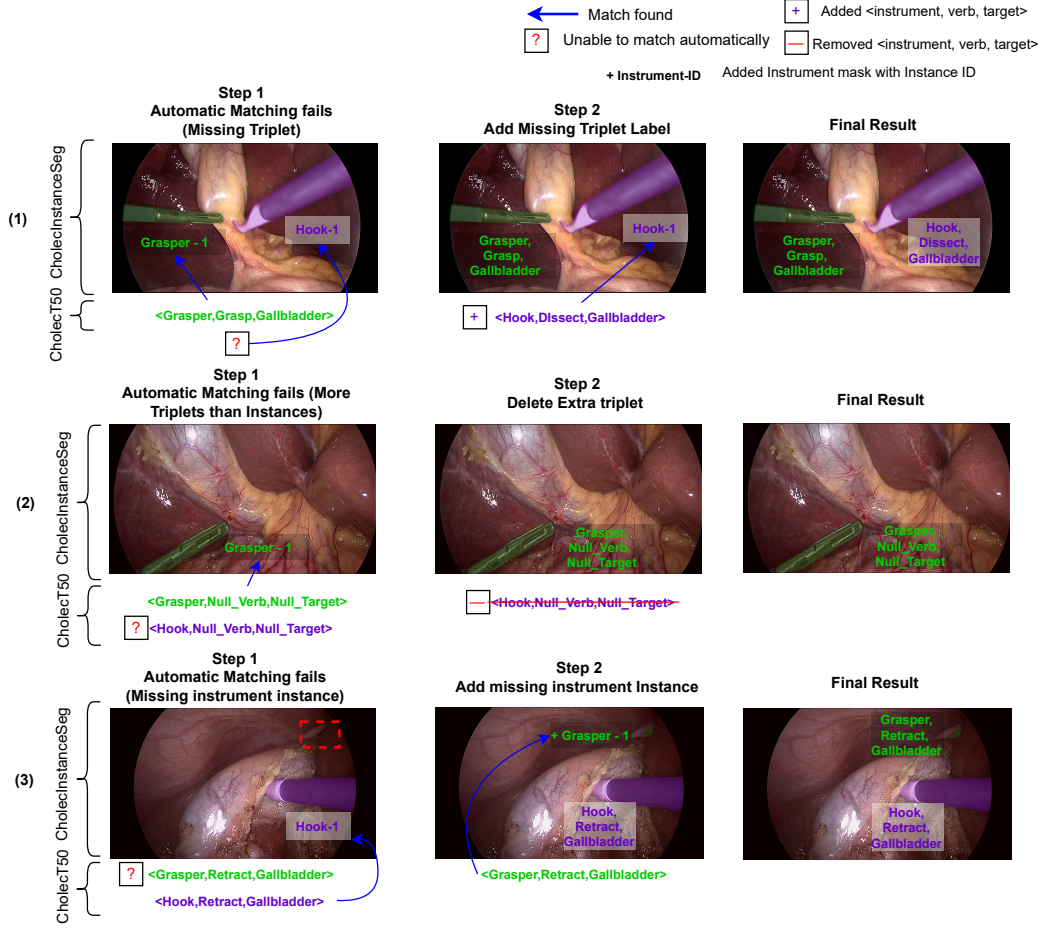


Figure 4.3: Examples of conflict errors between source annotations, including missing triplets, redundant triplets, and missing instrument masks.

4.4 Evaluation Protocol

Triplet segmentation requires evaluation of both semantic correctness and spatial grounding. This chapter therefore reports three related evaluation modes.

Recognition mAP evaluates whether triplets are present in a frame, regardless of spatial alignment. This corresponds to the standard frame-level classification setting used in CholecT50.

Detection mAP evaluates whether each predicted triplet is correctly associated with an instrument bounding box, using bounding-box IoU for localisation.

Segmentation mAP evaluates whether each predicted triplet is correctly associated with an instrument instance mask, using mask IoU for pixel-level grounding.

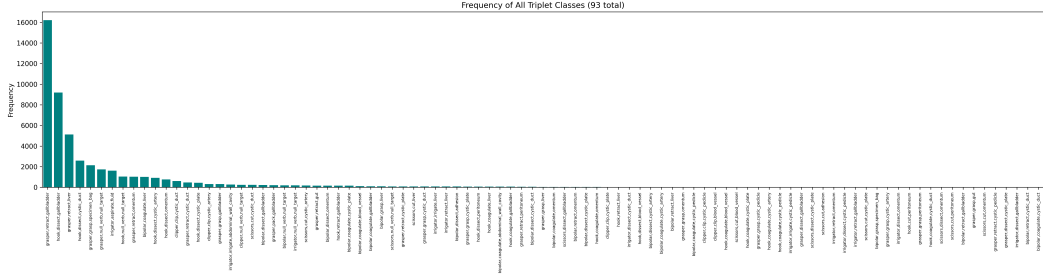


Figure 4.4: Frequency distribution of triplet classes in CholecTriplet-Seg. The dataset has a long-tailed distribution, with a small number of frequent triplets and many rare triplets.

Metrics are reported for individual components (mAP_I , mAP_V , mAP_T), component pairs (mAP_{IV} , mAP_{IT}), and full triplets (mAP_{IVT}). Superscripts indicate the evaluation mode, such as $\text{mAP}_{IVT}^{\text{seg}}$ for full triplet segmentation mAP. The full triplet score is the strictest measure because a prediction must correctly identify the instrument, verb, target, and spatial support.

4.5 TargetFusionNet

The main modelling challenge in triplet segmentation is target prediction. Instrument classes are visually distinctive and are directly tied to the mask. Verbs can often be inferred from tool pose and local motion cues, although this chapter uses single frames. Anatomical targets are harder: they may have weak boundaries, similar appearance, partial visibility, and ambiguous definitions. Pixel-level target annotations for the CholecTriplet-Seg target vocabulary are unavailable, and producing them would require extensive expert labelling.

TargetFusionNet addresses this bottleneck by incorporating weak anatomical priors into an instance segmentation architecture. The model builds on Mask2Former [Cheng 2022b], using transformer queries to predict masks and labels. It augments the decoder with a target-aware fusion mechanism that injects coarse anatomical information derived from EndoViT [Batić 2024], a tissue segmentation model trained on CholecSeg8k [Hong 2020a].

The anatomical priors are weak rather than definitive. EndoViT predicts broad tissue categories that only partially align with the CholecTriplet-Seg target vocabulary. For example, some outputs correspond closely to surgical targets, while others are approximate or cover multiple target categories. TargetFusionNet therefore treats these logits as noisy spatial context rather than as ground truth.

The target fusion mechanism has three parts:

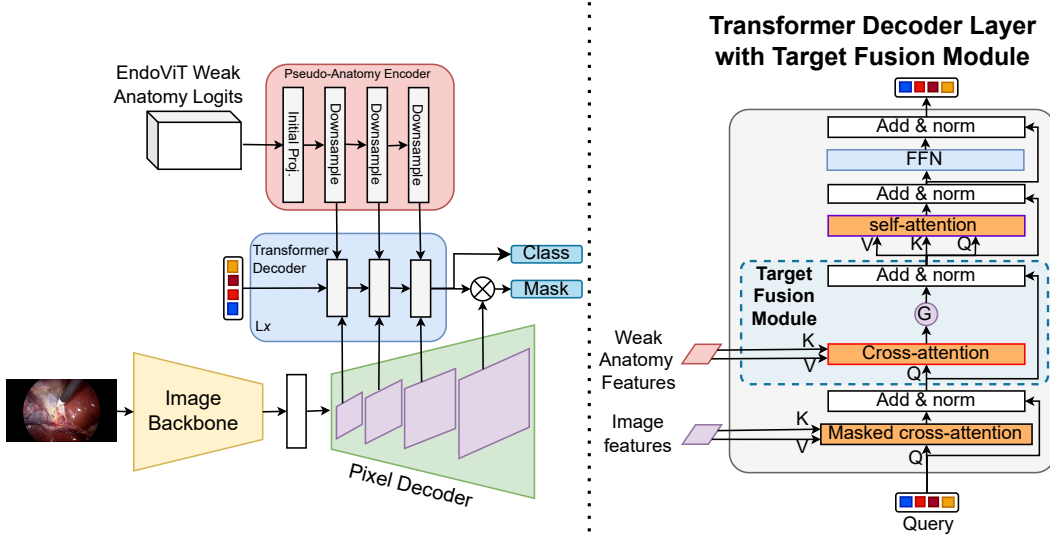


Figure 4.5: Overview of TargetFusionNet. Weak anatomical logits are encoded into multi-scale target-prior features and fused with Mask2Former decoder queries through gated cross-attention.

- **Weak-anatomy encoder:** EndoViT logits are projected into the decoder feature dimension and downsampled into multi-scale feature maps.
- **Target-aware decoder:** Mask2Former decoder layers are augmented so that instance queries can attend to the weak-anatomy features.
- **Gated cross-attention:** a learned gate controls how much anatomical context is injected into each query.

For a query Q and weak-anatomy key/value features (K_t, V_t) , the gated update is:

$$Q' = Q + \sigma(W_g A) \odot A, \quad A = \text{Attn}(Q, K_t, V_t),$$

where A is the cross-attention output, W_g is a learnable linear layer, σ is the sigmoid function, and $\odot$ denotes element-wise multiplication. The gate allows the network to use anatomical priors when they are helpful while suppressing them when they are noisy or misleading.

Each predicted instance mask receives a single label from the 100 clinically valid triplet classes. This 100-class formulation preserves triplet coherence: the network predicts a valid instrument-verb-target combination rather than three independent labels that may not form a meaningful triplet.

4.6 Experiments

TargetFusionNet was evaluated against three baselines:

- **RDV-Det**: a weakly grounded baseline derived from the Rendezvous framework, using class activation maps for coarse localisation [Nwoye 2023].
- **RDV + Mask2Former**: a hybrid baseline that replaces weak localisation with Mask2Former instance masks while retaining the RDV triplet association mechanism.
- **Mask2Former-Triplet**: a strong instance-level baseline that extends Mask2Former with a 100-class triplet head but does not include the target-aware fusion module.

TargetFusionNet was implemented in MMDetection [Chen 2019] with a ResNet-50 [He 2016] backbone pretrained for instrument segmentation. Images were resized to 1024×1024 , and training used AdamW [Loshchilov 2019] with random horizontal flipping, cropping, and scaling. Experiments were performed on a single NVIDIA A100 GPU, and the best checkpoint was selected using validation mask mAP.

4.7 Results

Table 4.3 reports performance across segmentation, detection, and recognition. The main metric is mAP_{IVT}^{seg} , since it evaluates complete triplet correctness with instance-mask grounding.

Evaluation	Method	mAP_I	mAP_V	mAP_T	mAP_{IV}	mAP_{IT}	mAP_{IVT}
Segmentation	RDV-Det	0.09	0.11	0.08	0.08	0.04	0.03
Segmentation	RDV + Mask2Former	48.11	32.51	16.29	14.40	11.38	8.73
Segmentation	Mask2Former-Triplet	65.24	45.61	20.75	23.03	16.47	12.23
Segmentation	TargetFusionNet	67.19	46.27	21.55	24.93	17.75	13.47
Detection	RDV-Det	3.28	2.86	0.29	1.43	0.77	0.55
Detection	RDV + Mask2Former	54.28	34.48	13.54	16.56	9.30	6.72
Detection	Mask2Former-Triplet	64.64	45.35	20.64	22.94	16.42	12.19
Detection	TargetFusionNet	66.52	46.23	23.70	24.82	17.74	13.45
Recognition	RDV-Det	80.55	61.44	40.83	35.93	34.47	27.17
Recognition	RDV + Mask2Former	80.55	61.44	40.83	35.93	34.47	27.17
Recognition	Mask2Former-Triplet	82.70	64.64	50.22	39.79	39.93	32.28
Recognition	TargetFusionNet	86.61	71.93	51.19	44.48	43.24	34.23

Table 4.3: Triplet segmentation, detection, and recognition results. Values are mAP scores for instruments, verbs, targets, pairs, and full triplets.

The results show three important trends. First, weak localisation is insufficient for instance-level triplet grounding: RDV-Det achieves only 0.03 mAP_{IVT}^{seg} . Second, adding strong instance masks provides a large improvement, raising mAP_{IVT}^{seg} to 8.73 in RDV + Mask2Former. Third, jointly predicting masks and triplet labels with Mask2Former-Triplet improves the score to 12.23, while TargetFusionNet reaches 13.47 by incorporating target-aware fusion.

The detection and recognition results show the same pattern. TargetFusionNet achieves the best full-triplet detection score, $\text{mAP}_{IVT}^{det} = 13.45$, and the best recognition score, $\text{mAP}_{IVT}^{rec} = 34.23$. The recognition results are higher than the segmentation scores because they do not require spatial grounding; however, the segmentation scores are the more relevant measure for this chapter’s goal.

4.7.1 Fusion Ablations

Table 4.4 compares TargetFusionNet with early and late fusion variants. Early fusion concatenates weak anatomical logits with the RGB image before the backbone. Late fusion injects the logits after the decoder. TargetFusionNet performs mid-level gated fusion inside the transformer decoder.

Method	mAP_I^{seg}	mAP_V^{seg}	mAP_T^{seg}	mAP_{IV}^{seg}	mAP_{IT}^{seg}	mAP_{IVT}^{seg}
TargetFusionNet	67.19	46.27	21.55	24.93	17.75	13.47
Early-concat	69.13	50.74	21.52	27.62	16.83	12.21
Late-concat	59.90	40.65	20.21	23.17	16.07	10.99

Table 4.4: Segmentation mAP for target-fusion strategies. Although early fusion improves some component scores, TargetFusionNet gives the best full-triplet segmentation score.

Early fusion improves some component scores, especially instrument and verb segmentation, but does not improve full triplet segmentation. This suggests that better component predictions do not necessarily produce coherent instrument-verb-target combinations. Late fusion performs worse, indicating that anatomical context is less useful when injected only after the decoder has already formed its instance representations. The gated decoder-level fusion in TargetFusionNet offers the best full triplet performance.

4.7.2 Qualitative Results

Figure 4.6 compares predictions from RDV-Det, RDV + Mask2Former, Mask2Former-Triplet, and TargetFusionNet. The examples show that all methods can perform reasonably in simple single-instrument cases, but differences emerge in complex

frames containing multiple instruments, occlusion, or rare targets. TargetFusion-Net more consistently links the correct instrument instance to the correct target, particularly in cases where weak anatomical priors help distinguish visually similar structures.

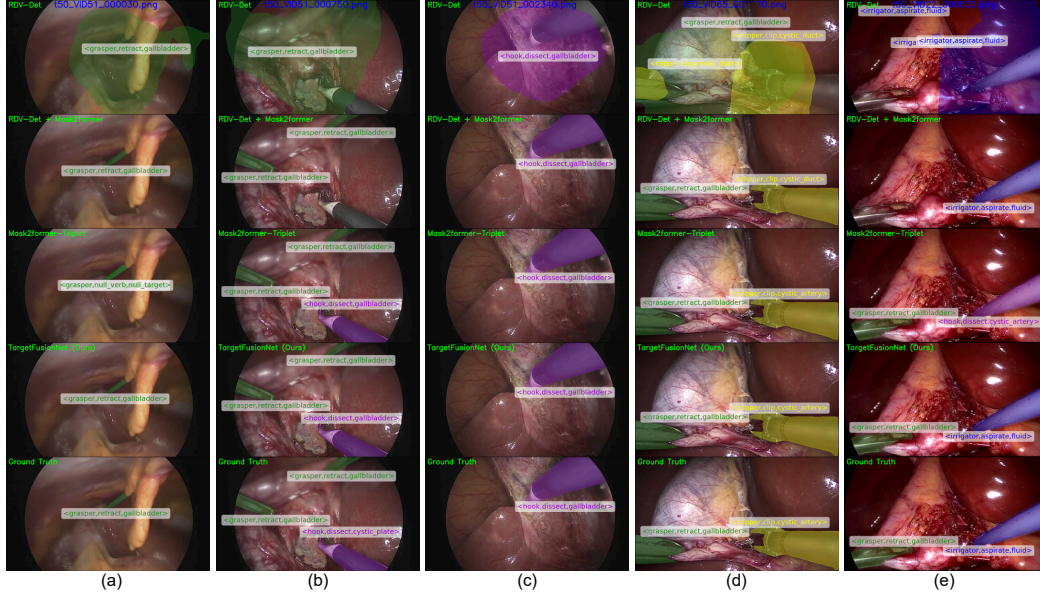


Figure 4.6: Qualitative comparison of grounded triplet predictions. TargetFusion-Net improves spatial grounding and triplet consistency in complex multi-instrument scenes.

4.8 Failure Modes and Limitations

Despite the improvements from strong instance supervision and target-aware fusion, triplet segmentation remains difficult. The most important limitation is target prediction. Anatomical targets can be visually similar, weakly bounded, partially occluded, or defined by surgical context rather than appearance alone. This is why target mAP remains much lower than instrument mAP across all methods.

Four recurring failure modes were observed. First, missing or incorrect instrument masks can cause the entire triplet prediction to fail, since triplet segmentation depends on accurate instance localisation. Second, rare triplet classes are difficult to learn because they have limited supervision. Third, visually similar targets such as cystic duct, cystic artery, cystic pedicle, and gallbladder are often confused. Fourth, some verbs are difficult to distinguish in single frames; for example, grasping and retracting may look similar without temporal context.

The long-tailed distribution of triplets further amplifies these issues. When

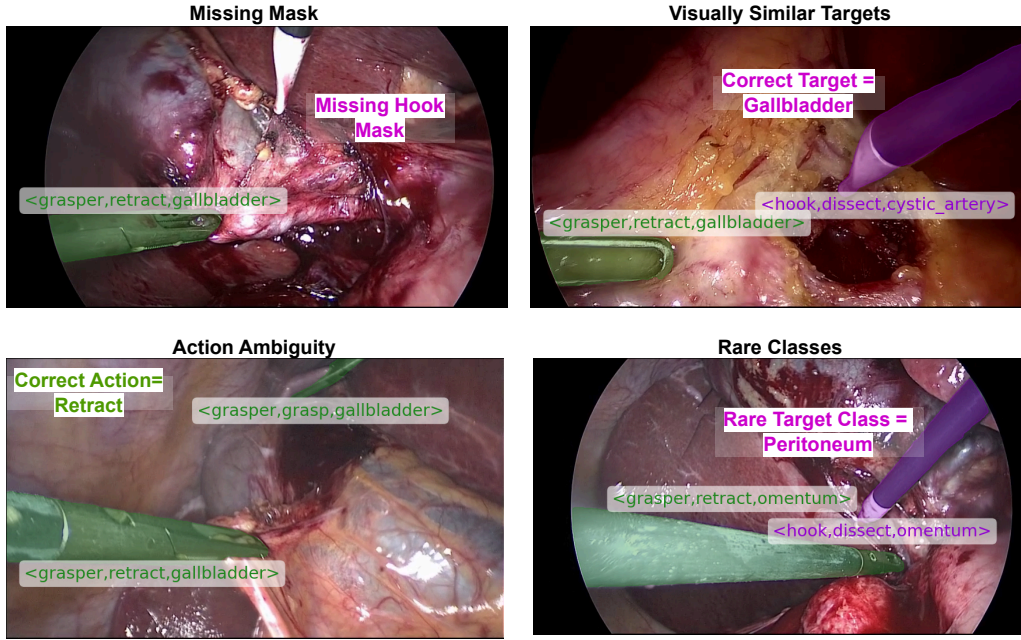


Figure 4.7: Representative triplet segmentation failure modes, including missing masks, confusion between visually similar anatomical targets, verb ambiguity, and errors on rare target classes.

triplet classes are grouped by frequency, performance decreases sharply for rare classes.

Triplet group	Classes	Avg. samples/class	mAP_{IVT}^{seg}
Frequent	10	4168.9	0.4148
Medium-frequency	21	307.9	0.2071
Rare	62	24.9	0.04

Table 4.5: Triplet segmentation performance grouped by training frequency. Rare triplets are substantially harder to predict.

These limitations motivate the next chapter. TargetFusionNet treats anatomical information as a weak visual prior and fuses it into a representation-learning model. However, many target errors appear to require reasoning about anatomical context, instrument position, and surgical intent. Chapter 5 therefore explores whether target prediction should be approached not only as a feature-fusion problem, but also as a reasoning problem.

4.9 Conclusion

This chapter introduced triplet segmentation as a spatially grounded formulation of surgical action triplet understanding. By aligning CholecT50 triplet labels with CholecInstanceSeg instrument masks, CholecTriplet-Seg provides the first benchmark in this thesis for instance-level grounding of instrument-verb-target interactions. TargetFusionNet improves over weakly grounded and strongly supervised baselines by incorporating weak anatomical priors through gated decoder-level fusion.

The results show that strong instrument instance supervision is essential for grounded surgical interaction understanding, but also that target prediction remains the main bottleneck. This observation provides the conceptual bridge to the next chapter, which investigates target prediction through explicit reasoning with vision-language models.

Target Reasoning with Vision-Language Models

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Source note. This chapter will be adapted from the ongoing Qwen / target reasoning manuscript. Figures and final results are expected to evolve, so this chapter is intentionally scaffolded more lightly than the previous mature paper chapters.

5.1 Chapter Overview

Drafting goal: frame target prediction as the persistent bottleneck in surgical action triplet understanding.

5.2 Why Target Prediction Is Difficult

Drafting goal: discuss anatomical ambiguity, visual similarity, occlusion, context dependence, and the limits of local visual representation.

5.3 From Target Fusion to Target Reasoning

Drafting goal: articulate the hypothesis that target prediction may require explicit reasoning over anatomy, instrument position, procedural context, and surgical intent.

5.4 Reasoning-Based Framework

Drafting goal: describe the pipeline: instrument instances, multimodal model input, weak anatomical priors, structured prompting, and final triplet prediction.

5.5 Target-Centric Chain-of-Thought

Drafting goal: define the target-centric reasoning procedure, including evidence gathering, ambiguity handling, alternative hypotheses, and final prediction.

5.6 Experiments

Drafting goal: leave space for final datasets, baselines, prompts, metrics, and result tables.

5.7 Discussion

Drafting goal: discuss what reasoning models reveal about surgical target prediction and how this reframes the earlier TargetFusionNet findings.

Discussion and Conclusion

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6.1 Summary of Contributions

Drafting goal: summarise the thesis arc from review, to dataset creation, to grounded triplet segmentation, to target reasoning.

6.2 Scientific and Technical Implications

Drafting goal: discuss what the thesis contributes to surgical data science, computer-assisted intervention, multitask learning, and interpretable surgical AI.

6.3 Limitations

Drafting goal: synthesise limitations across chapters: procedure specificity, annotation cost, long-tailed labels, anatomical ambiguity, evaluation constraints, and clinical translation.

6.4 Ethical and Clinical Considerations

Drafting goal: address responsible dataset release, privacy, bias, clinical deployment, and the difference between decision support and autonomy.

6.5 Future Work

Drafting goal: outline future directions including broader procedures, richer anatomical annotation, temporal reasoning, surgical foundation models, and prospective clinical validation.

6.6 Conclusion

Drafting goal: close by restating the thesis claim: spatial grounding and reasoning are necessary steps towards more useful surgical scene understanding.

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